Programming Fundamentals Using C++

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Week 03

Lecture – 01

Lecture – 02

Building Programs: Compiling and Linking

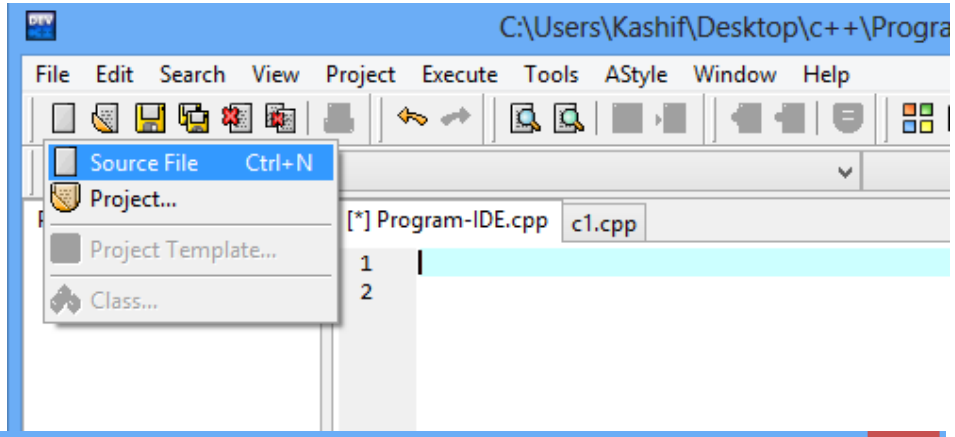
Semester Calendar

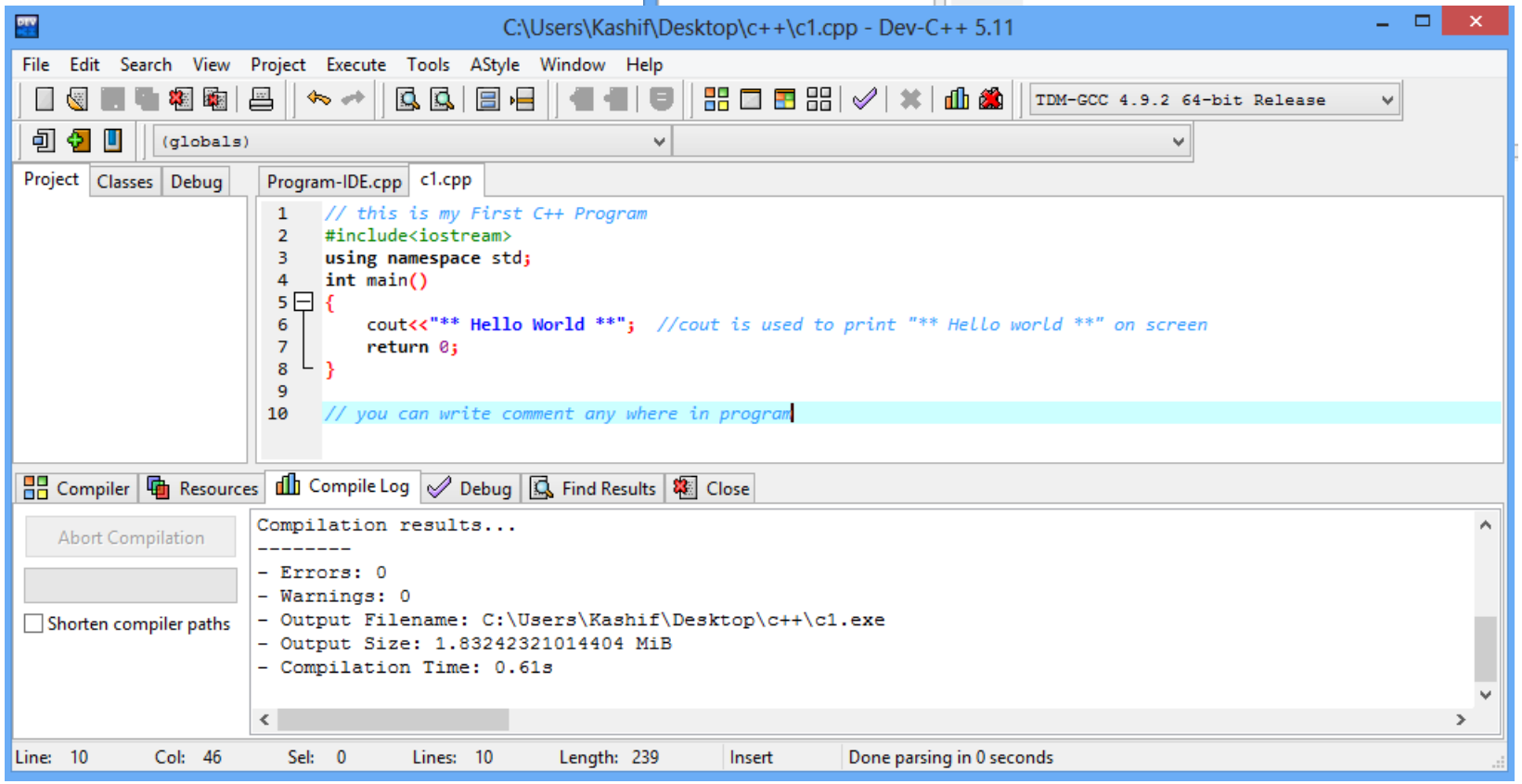
Class Schedule

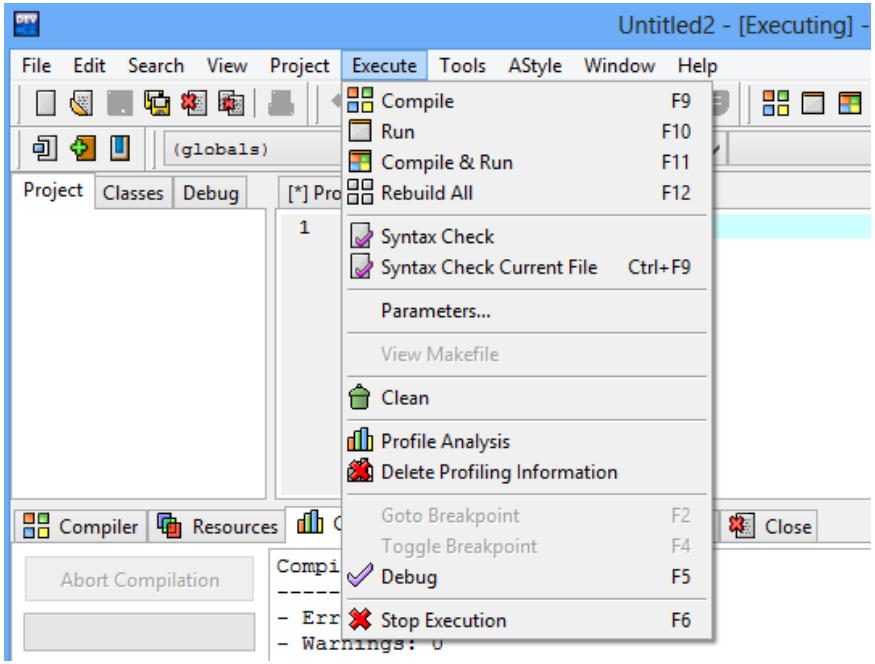
Lab Hours (Practice)

Tuesday (09:00 to 10:30)
Wednesday (09:30 to 12:30)
Wednesday (03:00 to 04:30)

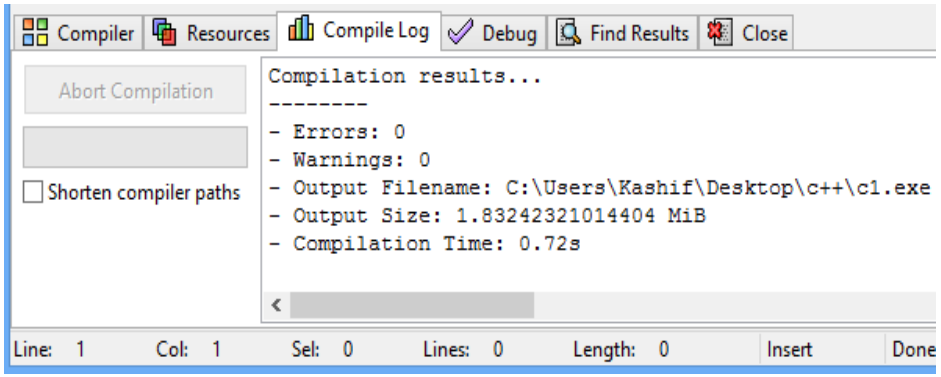
Creating 1st C++ program





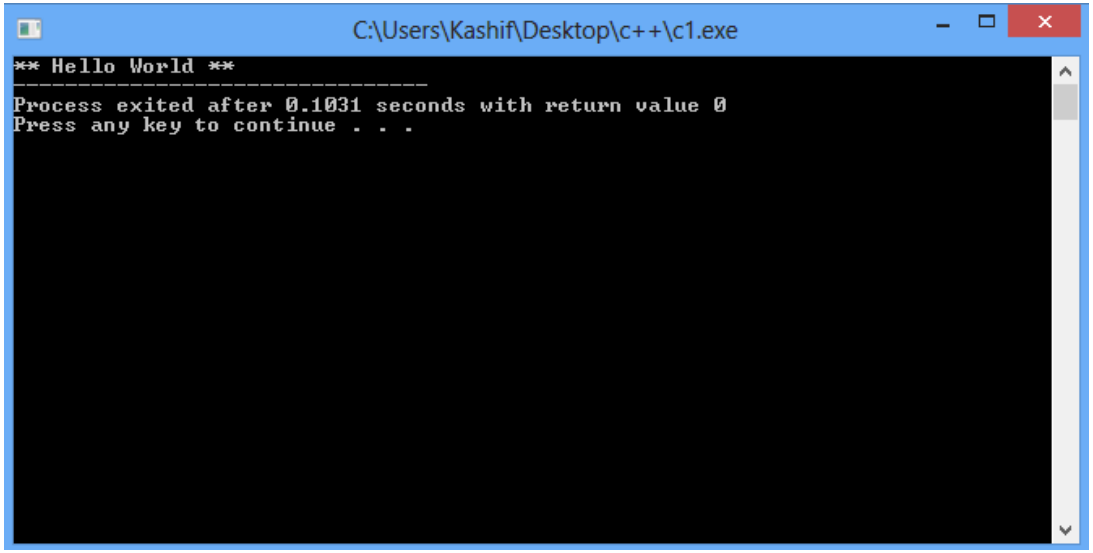


Build and Compile messages



Run, Compile & Rebuild Options

Console Window



Program Explained.

[*] c1.cpp

```
1   // this is my First C++ Program
2
3   #include<iostream>
4
5   using namespace std;
6
7   int main()
8
9   {
10      cout<<"** Hello World **";
11
12      return 0;
13  }
14
15
16
17
```

loads a *header* file containing function and class definitions

Loads a *namespace* called *std*. Namespaces are used to separate sections of code for programmer convenience. To save typing we'll always use this line

The *main* routine – the start of **every** C++ program! It returns an integer value to the operating system

The **return** statement returns an integer value to the operating system after completion. 0 means "no error". C++ programs **must** return an integer value.

- *cout* is the *object* that writes to the stdout device, i.e. the console window.
- It is part of the C++ standard library.
- Without the "using namespace std;" line this would have been called as *std::cout*. It is defined in the *iostream* header file.
- << is the C++ *insertion operator*. It is used to pass characters from the right to the object on the left.

Program statement vs preprocessor directive:

- Program statements are instructions to the computer to do something, such as adding two numbers or printing a sentence.
- A preprocessor directive, on the other hand, is an instruction to the compiler.
- A part of the compiler called the preprocessor deals with these directives before it begins the real compilation process.

Header Files:

- The preprocessor directive #include tells the compiler to insert another file into your source file.
- The type file usually included by #include is called a header file.
- The preprocessor directive #include tells the compiler to add the file iostream to the source file before compiling.

Why do this?

- iostream is an example of a header file.
- It's concerned with basic input/output operations, and contains declarations that are needed by the cout identifier and the << operator.
- Without these declarations, the compiler won't recognize cout and will think << is being used incorrectly

Always Start with main()

- When you run a C++ program, the first statement executed will be at the beginning of a function called main()
- The program may consist of many functions, but on startup, control always goes to main()
- If there is no function called main() in your program, an error will be reported when you run the program.
- The parentheses following the word main are the distinguishing feature of a function.
- Without parentheses compiler would think that main refers to some other program element.
- The word int preceding the function name indicates that this particular function has a return value of type int.

String Constants:

- The phrase in quotation marks, “** Hello World **”, is an example of a string constant.
- Its value is set when the program is written, and it retains this value throughout the program’s existence.
- The first statement tells the computer to display the quoted phrase.
- A semicolon signals the end of the statement.
- If you leave out the semicolon, the compiler will signal an error.

using namespace std;

Program 1

```
// My First C++ Program
#include<iostream>
using namespace std;
int main()
{
cout<<“ ** Hello World **”;

return 0;
}
```

Program 2

```
// My First C++ Program
#include<iostream>

int main()
{
std:: cout << "** Hello world **";

return 0;
}
```

Comments

- They help the person writing a program, and anyone else who must read the source file, understand what's going on.
- The compiler ignores comments
- Two styles of comments provided in c++
- comment starts with // and proceeds to end of line
- comment starts with /* and proceeds to first */

```
// This is an example of a comment.
/* This is another example of a comment. */
/* This is an example of a comment that
spans
multiple lines. */
```

Note:

comments of /* · · · */ style do not nest

```
/*
/* This sentence is part of a comment. */
This sentence is not part of any comment and
will probably cause a compile error.
*/
```

New Line

- They insert a new line.
- Both endl and \n serve the same purpose in C++

```
cout << endl      //  Inserts a new line
cout << "\n"      // Inserts a new line.
```

- Note: However, the key difference between them is that endl causes a flushing of the output buffer every time it is called, whereas \n does not

Any Question